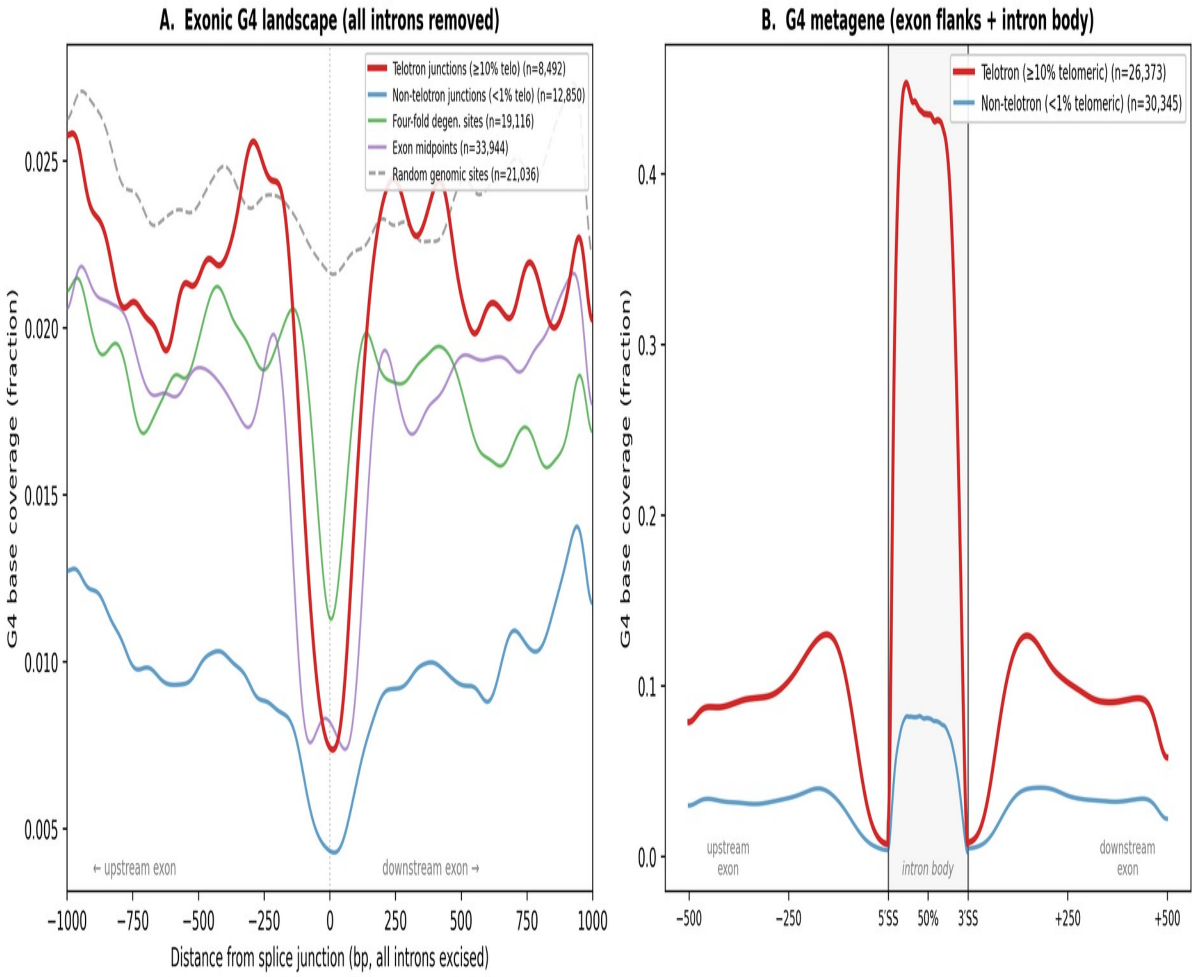
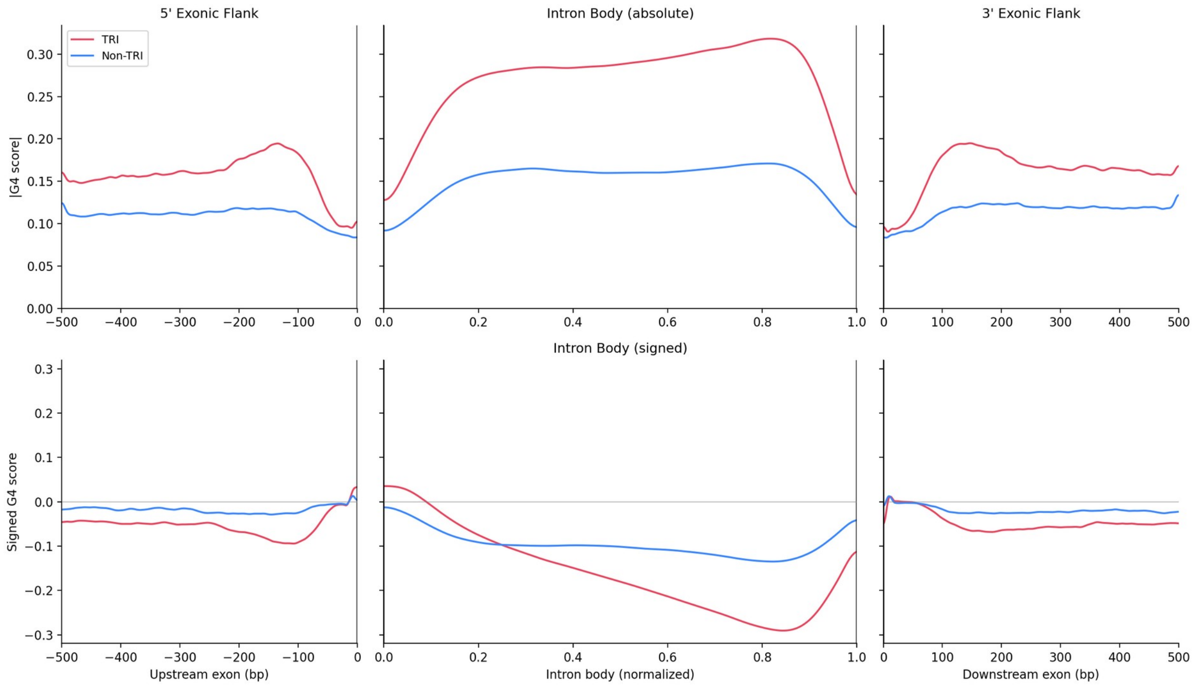
Extended Data Figures

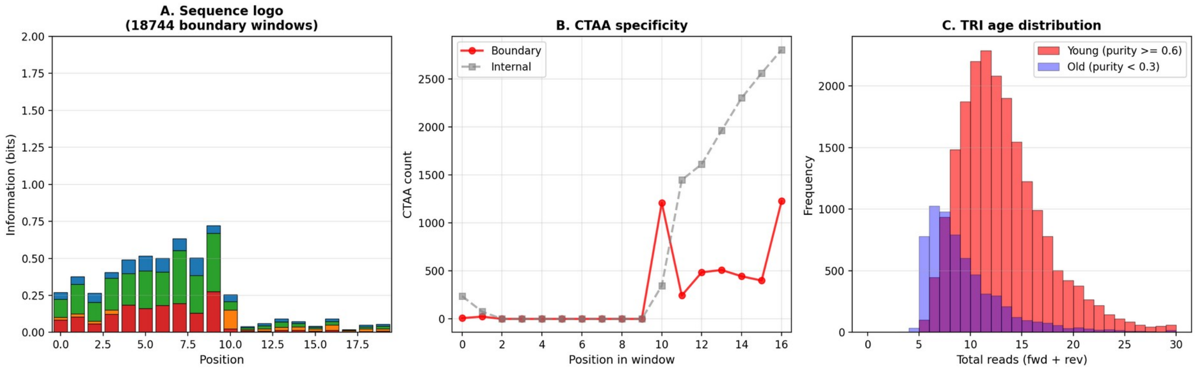
Extended Data Figure 1 | G-quadruplex landscape at telotron and non-telotron introns. (A) Per-base G4 motif coverage (G3+N1–7 × 4, both strands, Gaussian-smoothed σ = 30 bp) in pure exonic sequence flanking five classes of genomic locus. For splice-site loci, all introns in the surrounding gene were removed in silico to reconstruct the mRNA before extracting ±1 kb of exonic flanks. Introns were classified by telomeric hexamer content (≥10% = telotron, <1% = non-telotron), independent of splice dinucleotide. Telotron-flanking exons (red, mean 0.021) show ~2× higher G4 than non-telotron exons (blue, 0.010), suggesting a mild preference for G4-rich exonic context. (B) G4 metagene profile across 500-bp raw genomic flanks and the normalized intron body. Telotron intron bodies (red, mean 0.36) show 5.5× higher G4 than non-telotron bodies (blue, 0.06), consistent with TTAGGG/CCCTAA arrays forming G-quadruplex structures. Telomeric introns are found across all splice classes (~33% of GT–AG, CT–AC, and GC–AG introns alike), confirming that splice dinucleotide reflects strand orientation of the telomeric array, not a distinct insertion mechanism. Data from the eight Haptista MAGs with the most telotrons.



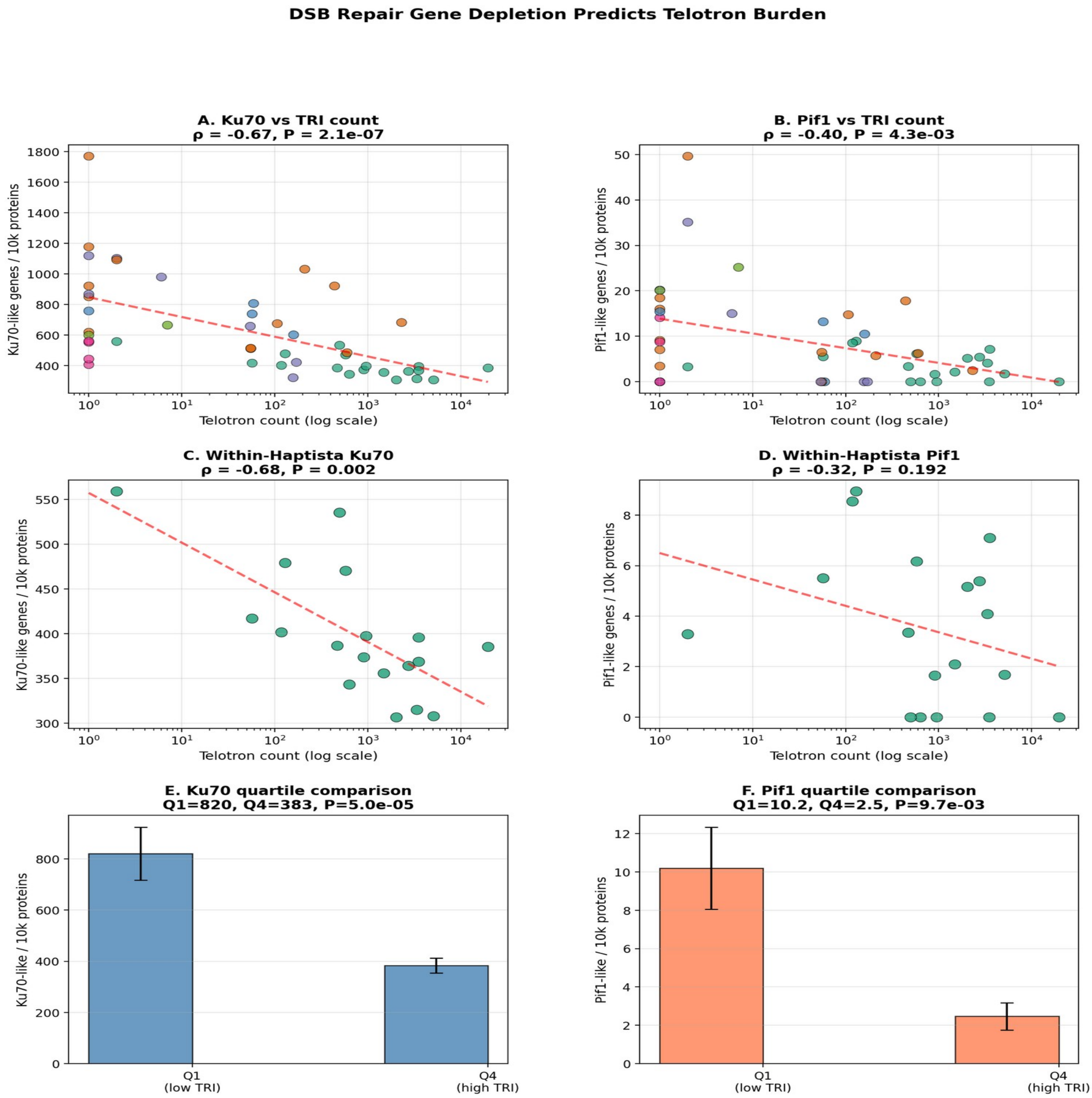
Extended Data Figure 2 | G4 metagene profile across telotrons and flanking exons. Top: absolute G4Hunter score (25-bp window) across telotrons (red) and non-telotron introns (blue), with 500-bp exonic flanks and normalized intron body. Telotrons show ~3-fold elevation. Bottom: signed G4 score reveals strand polarity within the intron body.



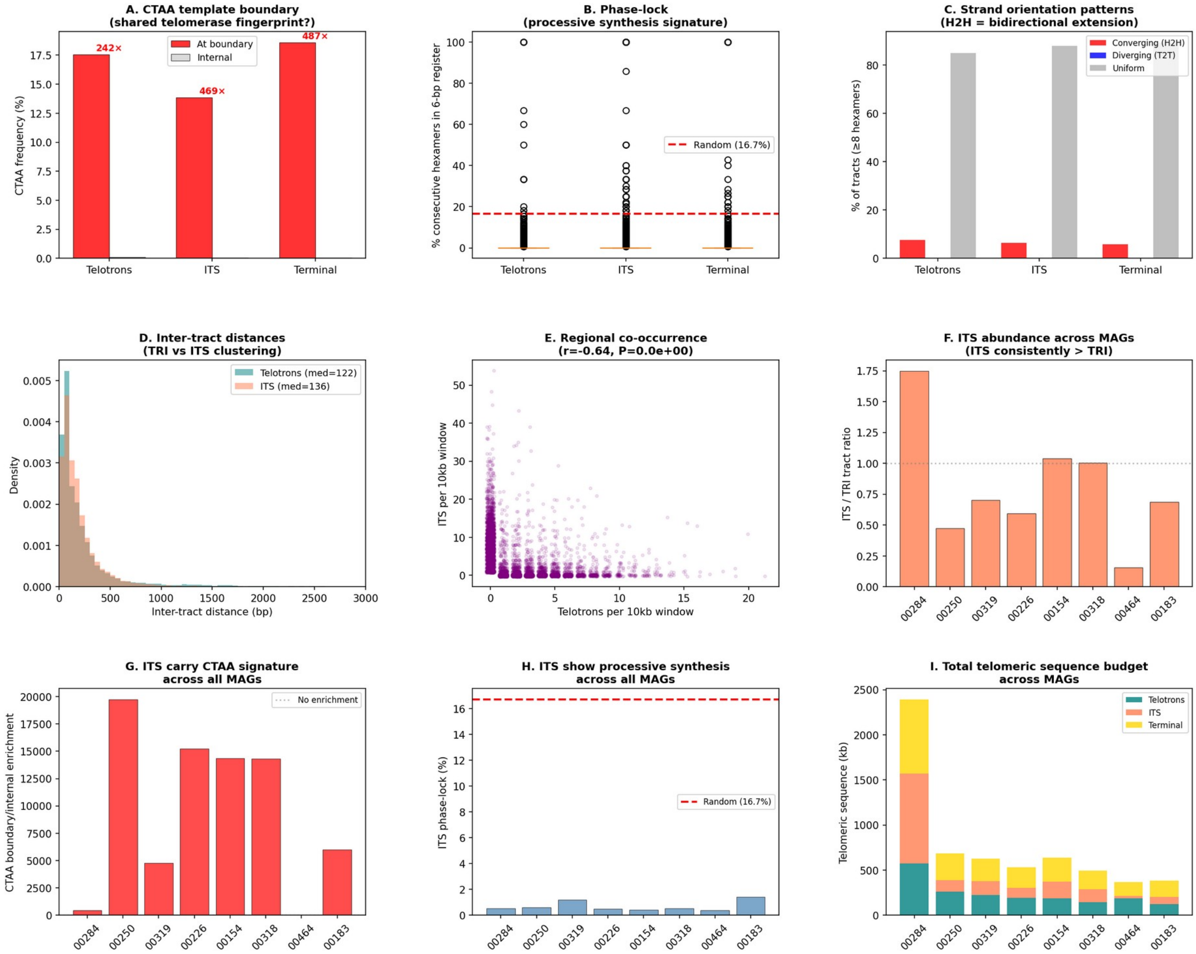
Extended Data Figure 3 | Sequence logos, CTAA specificity, and telotron age. Left: sequence logo at the 3' G-rich array boundary showing the TTAGGG repeat terminating at a defined position. Centre: CTAA tetranucleotide frequency is sharply enriched at the array boundary versus internal positions. Right: read-count distributions by telotron age; younger telotrons (high purity) have higher expression.



Extended Data Figure 4 | DSB repair gene depletion — extended analysis. (a,b) Ku70 and Pif1 vs telotron count across all MAGs. (c,d) Within-Haptista correlations controlling for phylogeny. (e,f) Top vs bottom quartile comparisons with error bars.



Extended Data Figure 5 | Interstitial telomeric sequence (ITS) analysis. (a) CTAA boundary comparison: telotrons, ITS, and terminal tracts all carry the same template-boundary signature. (b) Phase-lock. (c) Strand orientation patterns. (d) Inter-tract distances. (e) Regional co-occurrence. (f) ITS abundance across MAGs. (g) Cross-MAG CTAA in ITS. (h) Cross-MAG ITS phase-lock. (i) Telomeric sequence budget stacked bars.



Extended Data Figure 6 | Telomeric coverage spectrum and template-strand bias. Top: histogram of telomeric base coverage across all 51,532 tandem-6bp introns (log scale), with Haptista introns (dark blue, n = 46,121) overlaid on all lineages (light blue). The log-scale y-axis reveals a continuous bell-shaped distribution of telomeric content centered around 50–60%, invisible in the linear-scale view. Bottom: template-strand bias as a function of telomeric content among Haptista introns, showing a plateau at ~80% from 10% coverage onward, consistent with telomerase-mediated insertion on the template strand.